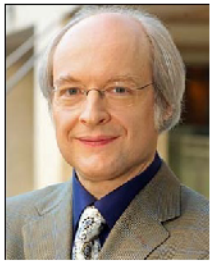
NILESEN 10 HEURISTICS

Introduction

Jakob Nielsen* (working along with Molich in 1990) proposed a set of ten guidelines that can be used as Principles of Design for a new Interface.

These also can be used as Heuristics for evaluating an Interface.

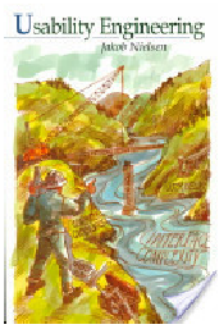
Since the ten guidelines were more in the spirit of rules of thumbs than specific rules, they are called as ‘ Heuristics’ rather than rules or laws that hold true in every case.



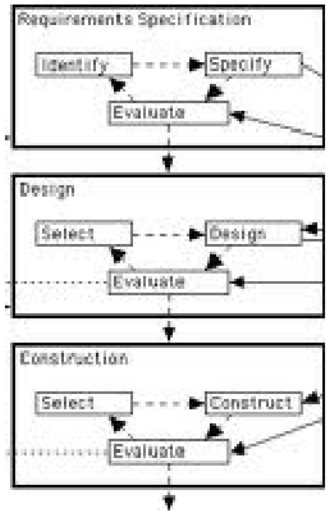
Jakob Nielsen

Usability consultant

Jakob Nielsen is a leading web usability consultant. He holds a Ph.D. in human–computer interaction from the Technical University of Denmark in Copenhagen.

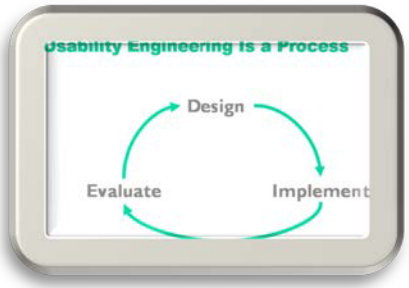


He has authored many books in Usability, HCI, & Experience design. His book titled " Usability Engineering" 1993 is a textbook on methods to make interfaces easier to use.



Usability Engineering

Usability Engineering involves User Research; Design Research and Validation of Design through Construction & User Testing.
In some institutions it is taught as an independent discipline while in others it is part of HCI discipline.



Introduction

Heuristics means “rules of thumb”.

These ten ‘rules of the thumb’ were derived after careful research by Nielsen who after conducting a factor analysis of 249 usability problems, came up with ten simply stated guidelines in 1994.

Nielsen’s heuristics method are empirically based derivations. Widely used by Usability professionals (which includes Interface designers), they are a means of quickly identifying likely interface design problems in an application.

Method suggested by Nielsen is popular because of its simplicity and low cost. It is preferred evaluation technique at the preliminary design stages by HCI professional.

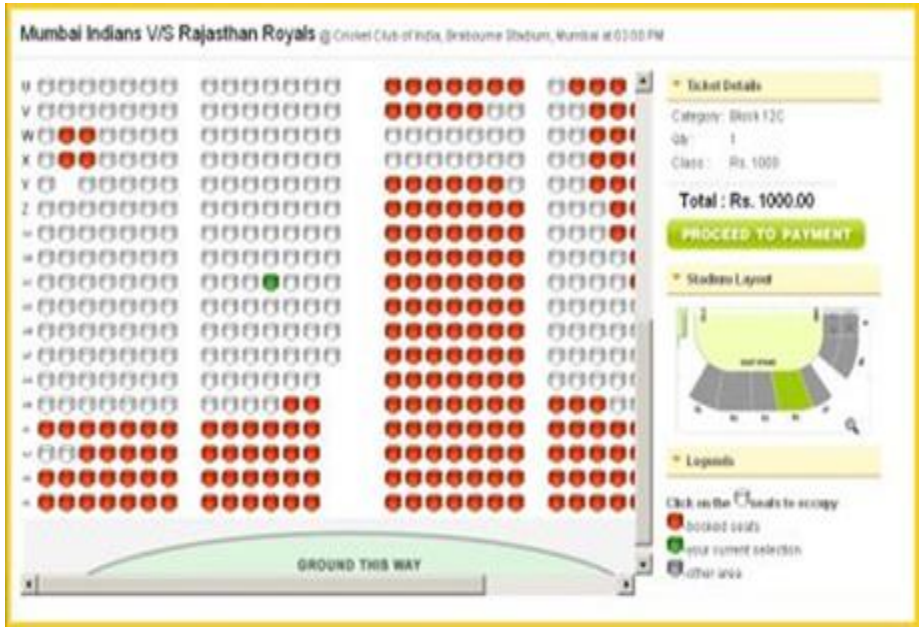
The Ten Heuristics as put forth by Nielsen.

1. **Visibility of system status**
2. **Match between system and the real world**
3. **User control and freedom**
4. **Consistency and standards**
5. **Error prevention**
6. **Recognition rather than recall**
7. **Flexibility and efficiency of use**
8. **Aesthetic and minimalist design**
9. **Help users recognize, diagnose, and recover from errors**
10. **Provision of Help and documentation**

Each principle is explained in consequent slides.

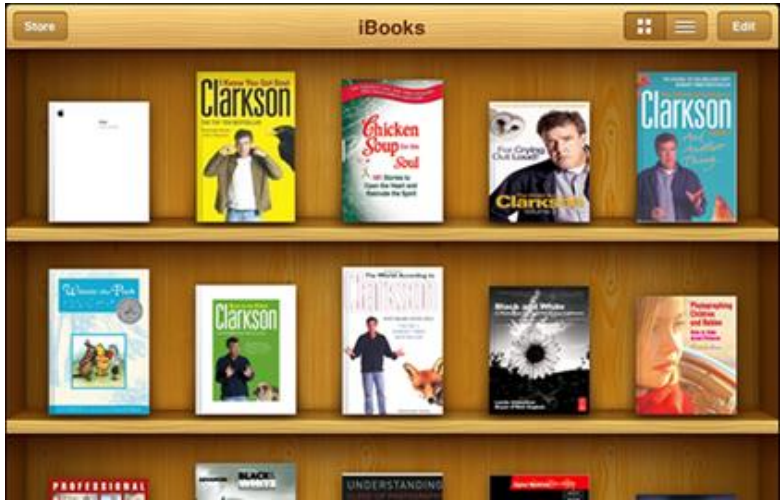
1. Visibility of system status

- The system should always keep users informed about what is going on, through appropriate feedback within reasonable time.



2. Match between system and the real world (use metaphor)

- The system should speak the users' language, with words, phrases and concepts familiar to the user, rather than system-oriented terms. Follow real-world conventions, making information appear in a natural and logical order.



3. User control and freedom

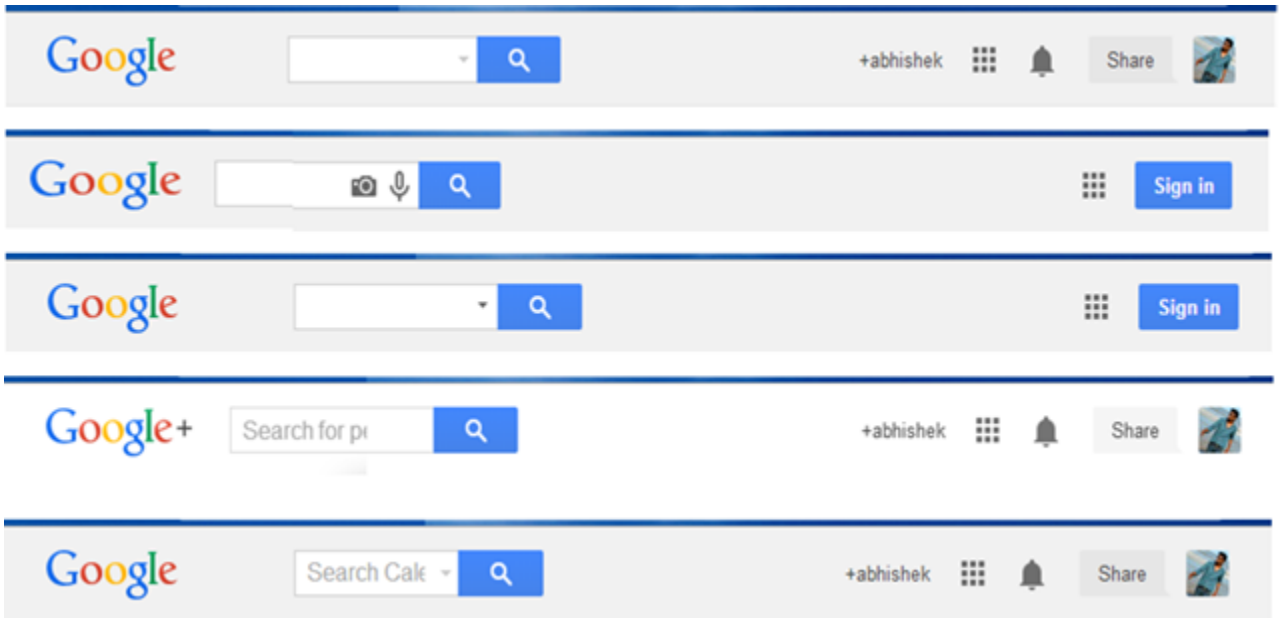
- Users often choose system functions by mistake and will need a clearly marked "emergency exit" to leave the unwanted state without having to go through an extended dialogue. Support undo and redo.

Ever felt the need for an undo button after sending an email to the wrong person?

It's a good thing Gmail allows that.

4. Consistency and standards

- Users should not have to wonder whether different words, situations, or actions mean the same thing. Follow platform conventions.



Consistency and standards (cont.)

- Consistency in Microsoft word and excel sheet

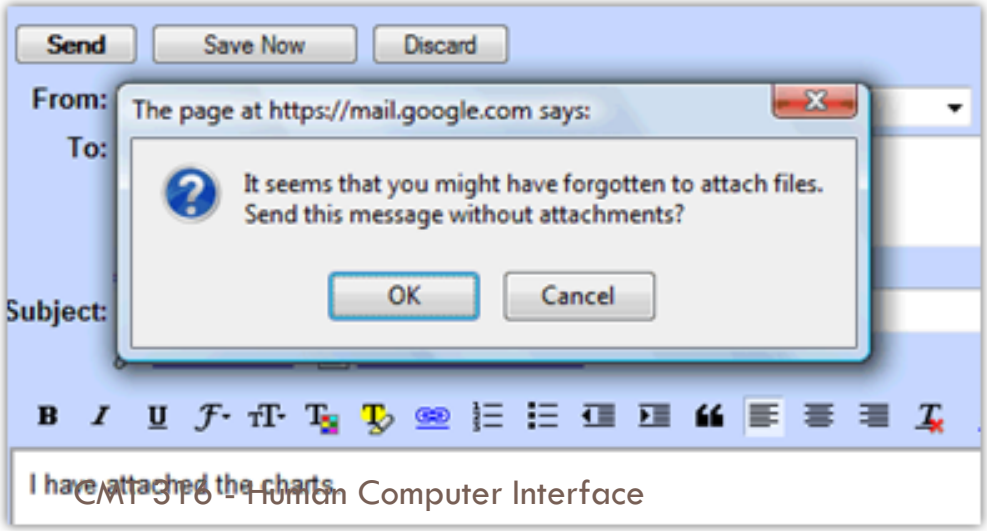
4.1 Microsoft Office

Word, Excel, and PowerPoint all use the same style toolbar with the same primary menu options: Home, Insert, Page Layout... Consistency results in efficiency and perceived intuitiveness.

Consistency and standards (cont.)

5. Error prevention

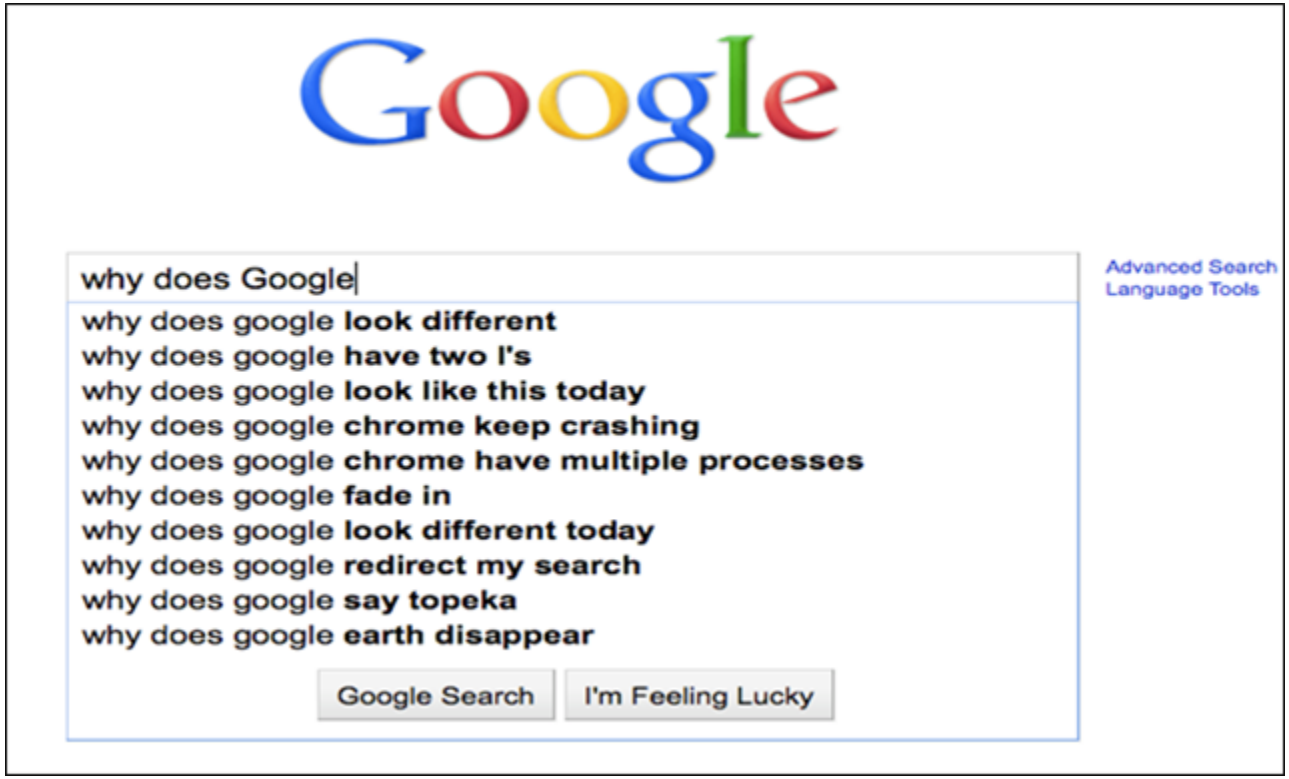
- Even better than good error messages is a careful design which prevents a problem from occurring in the first place. Either eliminate error-prone conditions or check for them and present users with a confirmation option before they commit to the action.



6. Recognition rather than recall

- Minimize the user's memory load by making objects, actions, and options visible. The user should not have to remember information from one part of the dialogue to another. Instructions for use of the system should be visible or easily retrievable whenever appropriate.

Recognition rather than recall (cont.)



Recognition rather than recall (cont.)

- Where is the shutdown option?

Important actions should be easy to access and identify.

In an earlier version of Windows 8, it was almost impossible to shut down the computer without googling how to do it.

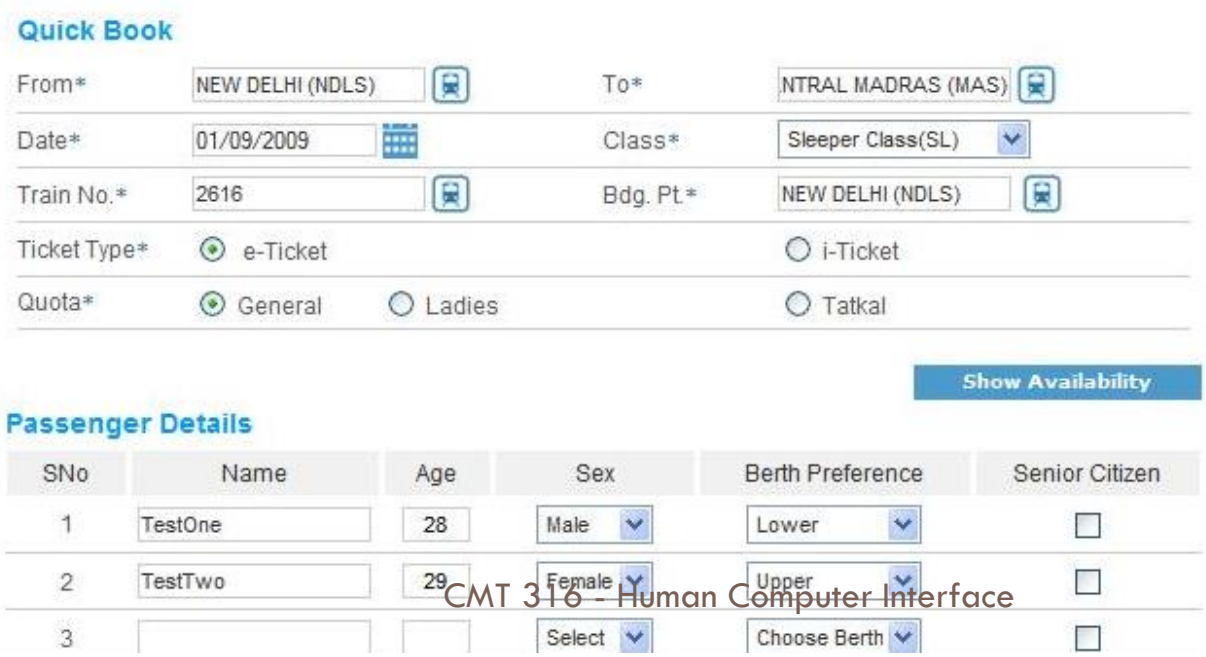
How To Shutdown Windows 8

1. Close all desktop apps you have opened.

2. Swipe in from the right edge of the screen, and then tap Settings. If you're using a mouse, point to the upper-right corner of the screen, move the mouse pointer down, and then click Settings.

7. Flexibility and efficiency of use

- Accelerators -- unseen by the novice user -- may often speed up the interaction for the expert user such that the system can cater to both inexperienced and experienced.,

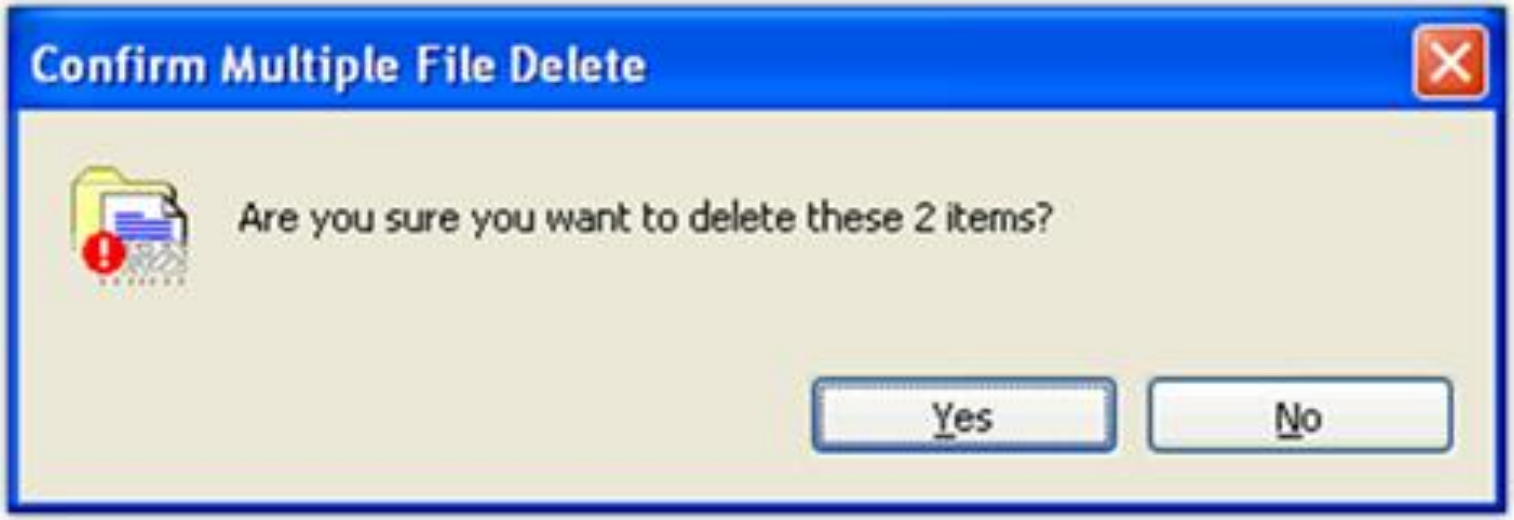


8. Aesthetic and minimalist design

- Dialogues should not contain information which is irrelevant or rarely needed. Every extra unit of information in a dialogue competes with the relevant units of information and diminishes their relative visibility.

9. Help users recognize, diagnose, and recover from errors

- Error messages should be expressed in plain language (no codes), precisely indicate the problem, and constructively suggest a solution.

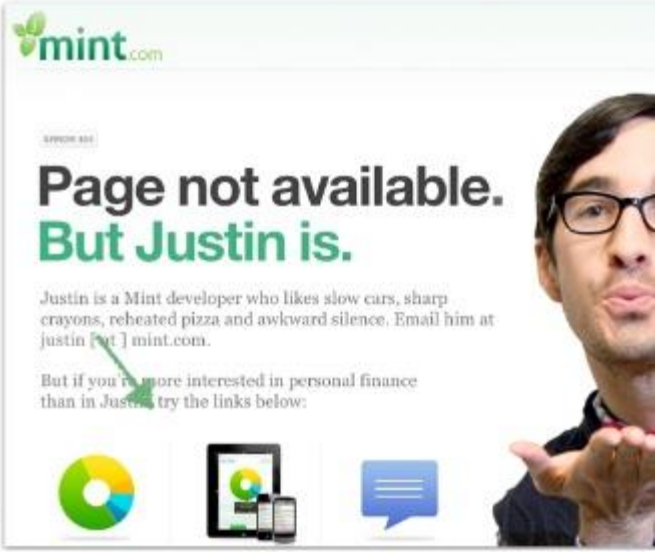


Help users recognize, diagnose, and recover from errors (cont.)

- It helps the user to diagnose something

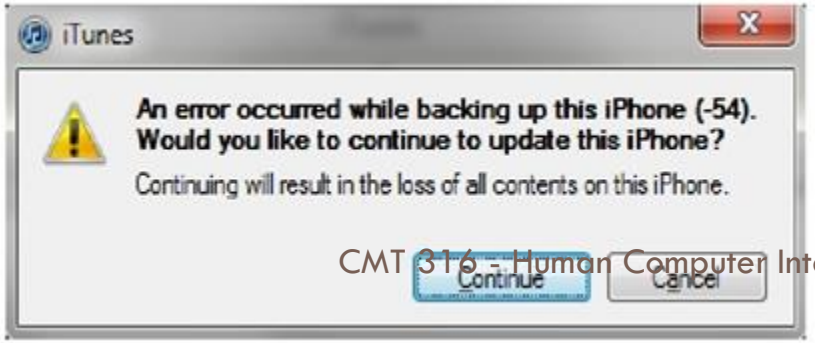
An assuring error message on Dropbox

A funny message keeps the audience engaged, while relevant links make sure they stay on your website.



Help users recognize, diagnose, and recover from errors

- Oops! What is the error here?
- No recovery information



10. **Help and documentation**

- Even though it is better if the system can be used without documentation, it may be necessary to provide help and documentation. Any such information should be easy to search, focused on the user's task, list concrete steps to be carried out, and not be too large.

Conclusions

- These 10 heuristics of usability help in refining a potential design into a good design. They ensure that interfaces evolve in the right direction
- These rules of the thumb act a check list to evaluate a design
- They also can be used as check list while evaluating any GUI

EVALUATION TECHNIQUE

Heuristic Evaluation

Heuristic Evaluation

- Heuristic evaluation is a systematic process of inspecting of a user interface
  - For usability problems
- It is both
  - Before design finalization- Predictive method
  - After design completion – Evaluation and rating method

Heuristic Evaluation (contd.)

- Heuristic evaluation method involves having a small set of evaluators (5 to 7) examine the interface and judge its compliance
  - Usability principles
  - Such as nelson's 10 usability principles

Heuristic Evaluation (contd.)

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- An example of how heuristics analysis is conducted is presented here.

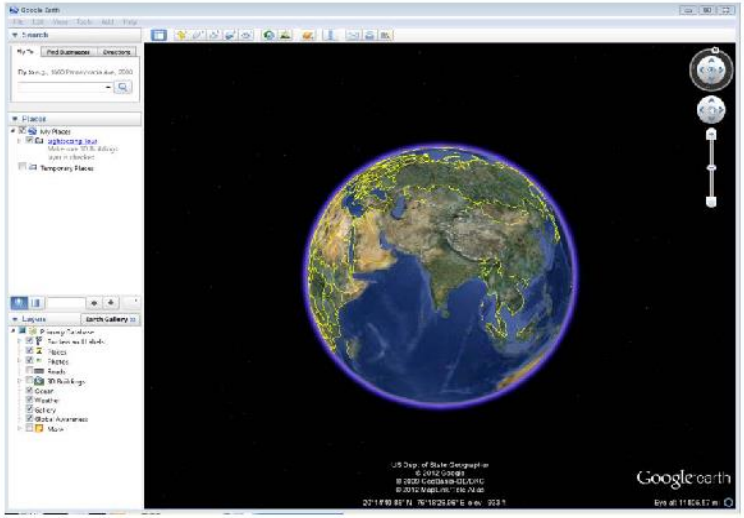
Case Study 1 **GOOGLE MAPS in Goggle Earth**

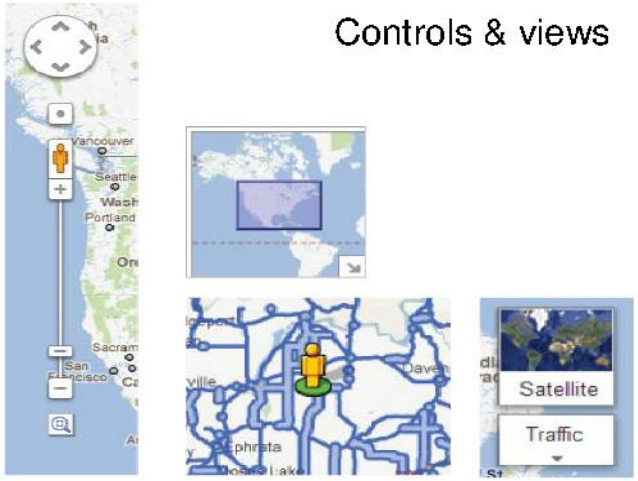
Evaluator: 1 nos. Expert user

Heuristics Used: Nielsen's Ten Heuristics

Google Maps is a well known free service provided by Google world wide.

It's not just a bunch of maps, it includes multiple layers: roads, terrain, satellite, street view, traffic etc. It also integrates user ratings and pictures with locations and businesses in the area.

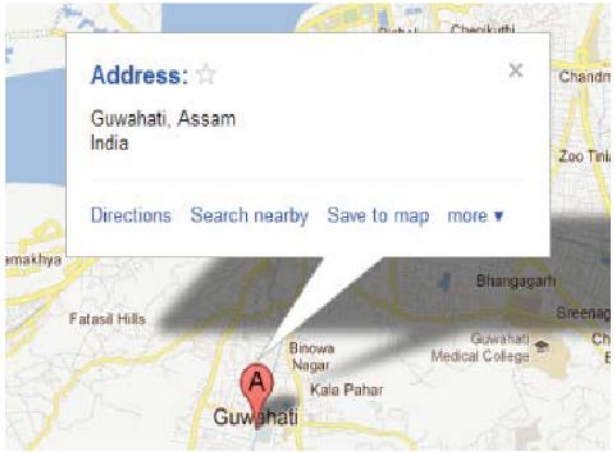


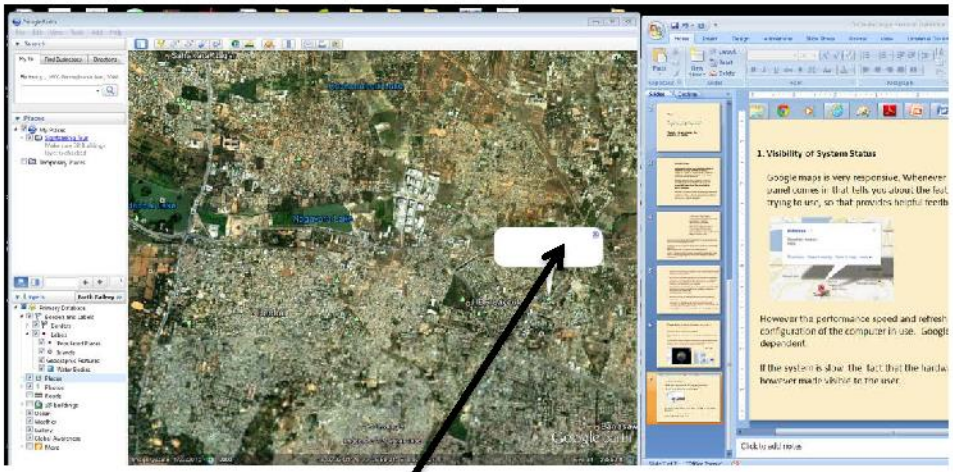


1. Visibility of System Status

Findings of the expert on visibility & status capability of Google earth is explained bellow

"Google maps is very responsive. Whenever you click a button, a panel comes in that tells you about the feature that you're trying to use, so that provides helpful feedback."





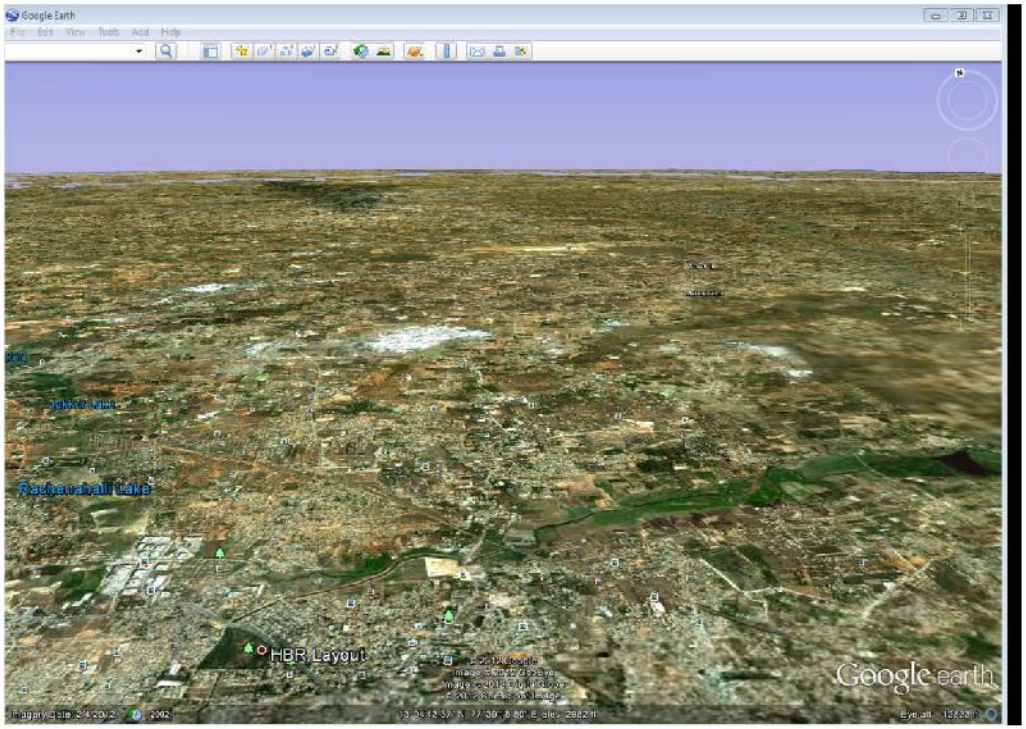
"System status if the Network connection is absent or suddenly is lost is not made visible to the user. The user sees a blank label" (second screen shot)

"However the performance speed and refresh rate depends on the hardware configuration of this computer in use. Google is very CPU intensive and RAM dependent.

If the system is slow due to hardware mismatch this fact that the hardware is not optimum - is not however made visible to the user."

2. Match between System and Real World

Real world



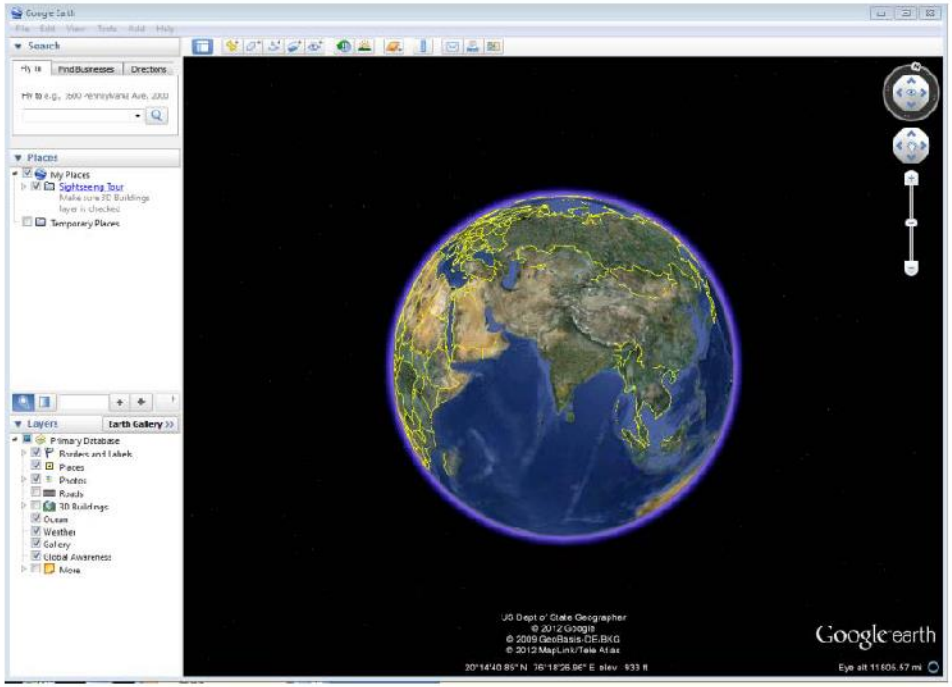
system

Closeness to real world is very good.

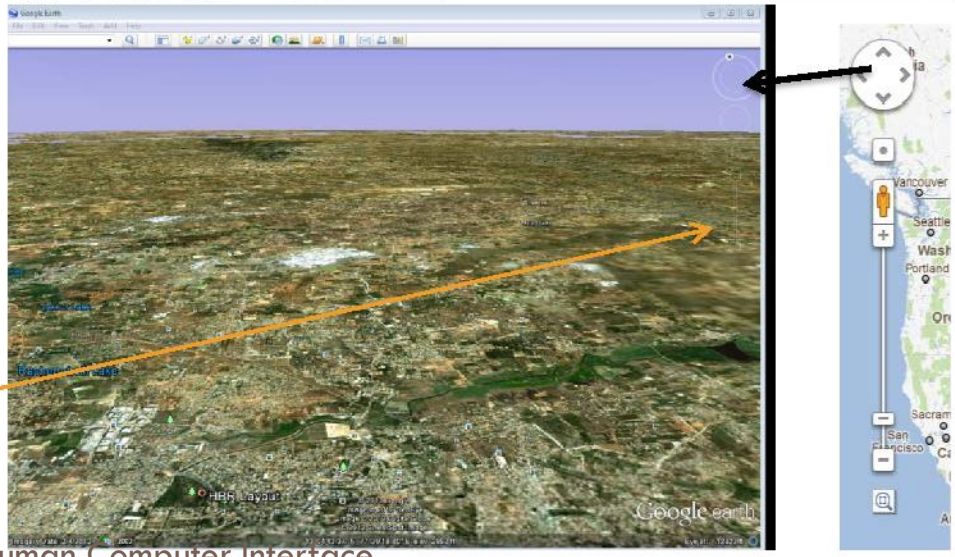
The window can be panned and horizon can be lowered giving a very Real world view.

3. User Control and Freedom

Almost all the features are available as checkboxes. When they are checked, those items are added to the map and when they're not checked they go right back off the map.



The zoom in controls are fairly intuitive. They recede into the background and come alive when mouse hovers on it. The direction ring gives full control to the user.



Freedom for the experienced user but for a novice a disappearing zoom slider bar can be confusing!

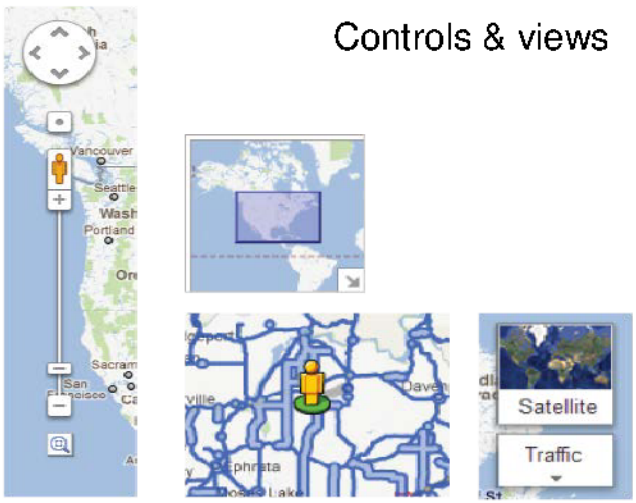
4. Consistency and Standards

Google maps is pretty consistent with the words and phrases that they use.

the symbols are pretty clear and a user could probably figure it out without even needing the labels.

Successive screens maintain consistency in continuity in terms of how & where tools / buttons appear.

The response of various interactions is standardized across the entire application.

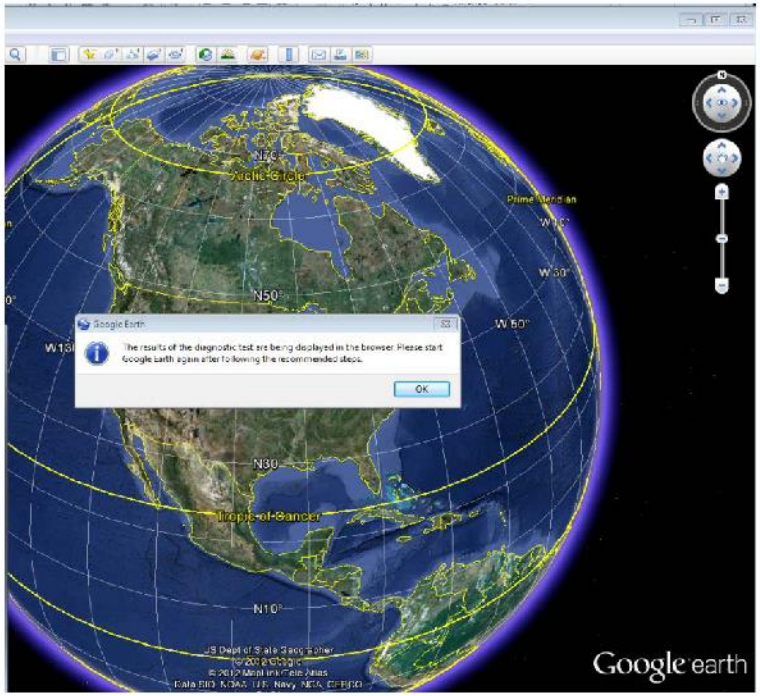


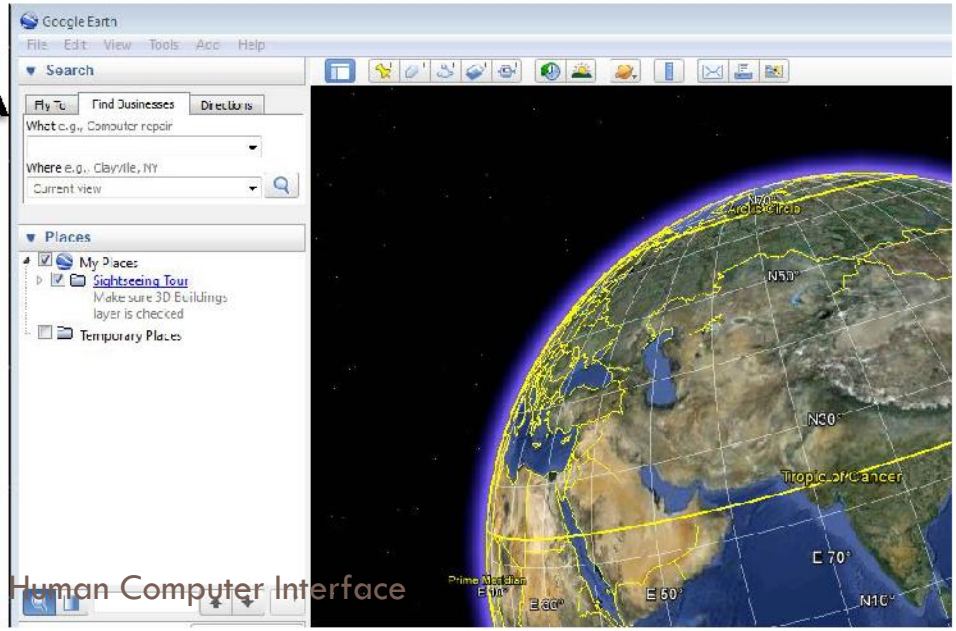
5. Error Prevention

There really isn't much error involved in a mapping program.

There is a possibility that a user may enter the wrong address, but Google maps is well configured to automatically decipher what the user may have been looking for and presents them with a list of options that could be correct.

This not only prevents accidental errors but also suggest corrections for the user.



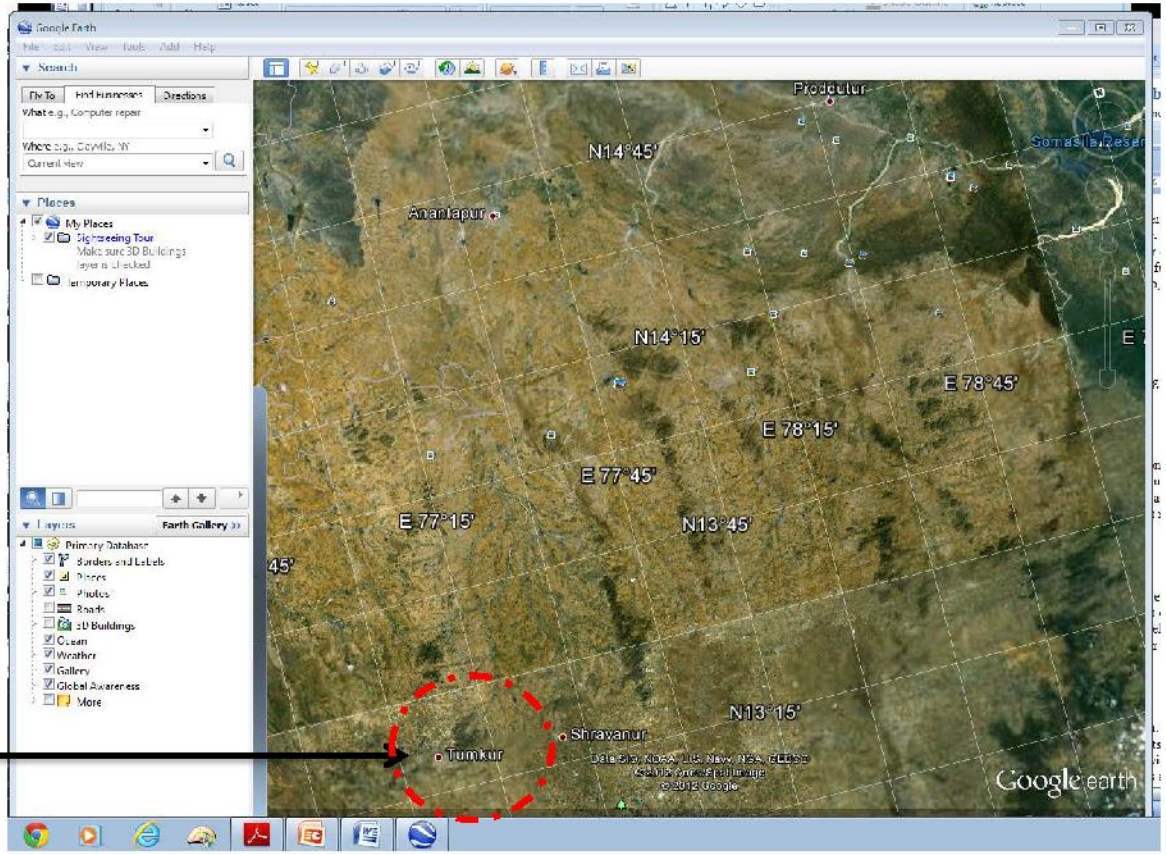


6. Recognition Rather than Recall

While most of the more useful functions are visible, but a user must click a button to get directions.

Navigating by panning and zooming often leads to being lost . Users often want to know which direction they need to pan.

Ex: On the right TUMKUR is visible. For a user to now go to BANGALURU & (if the user is not sure where Bangalore is with respect to Tumkur - East, West, North, South) or is not familiar with the latitude - longitude of Bangalore – will have to recall or recognise or resort to trail and error by first zooming out. Zooming out also leads to disappearing of small towns like Tumkur -. At lower zoom levels the user will have to 'RECAL' which is not an easy thing to do on a map of an unfamiliar geography.



It is here that recall is required or in order to become aware (recognize) where in the map one is with respect to overall map.

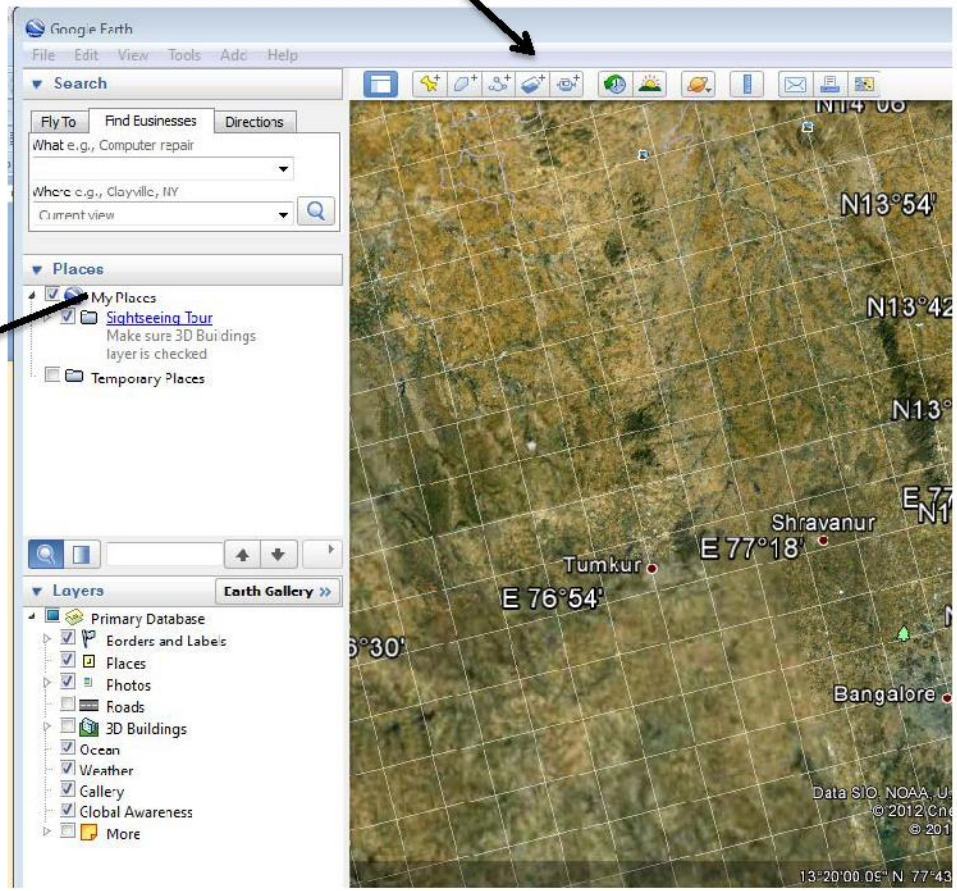
7. Flexibility and Efficiency of Use

The buttons on the task bar reflect the flexibility that is built in.

The Google Earth is highly flexible & efficient. Even if a GPS connection is not available

A user can set a reference location " starting point" and also key in where one wants to go –

The software map navigates itself by either panning , zooming, turning, and makes the destination visible.

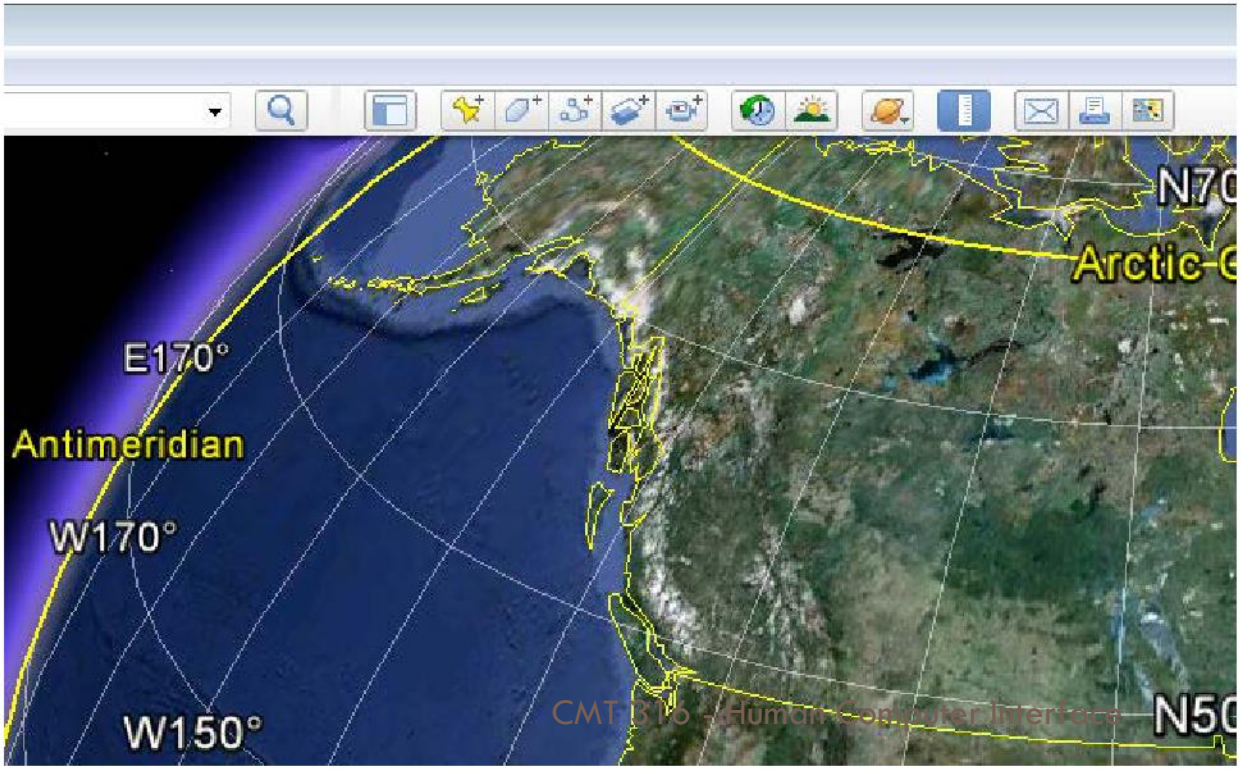


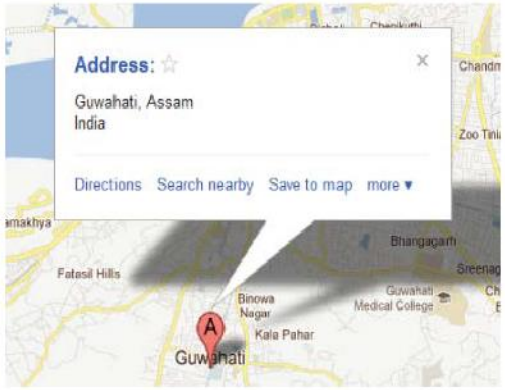
8. Aesthetic and Minimalist Design

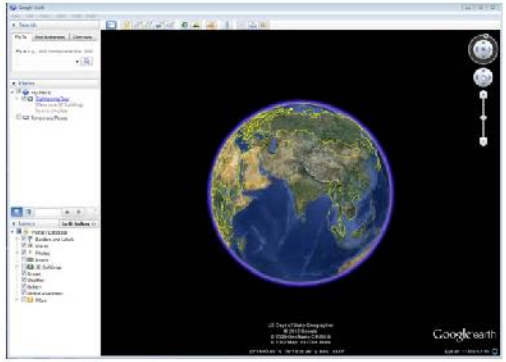
The entire user experience is pleasant. The design of the interactive buttons is subdued and minimalist.

The colour tone of the labels, instructions etc that appear are not loud and brash.

The whole graphics is tuned to make the Map window important (which is the main function of the software)

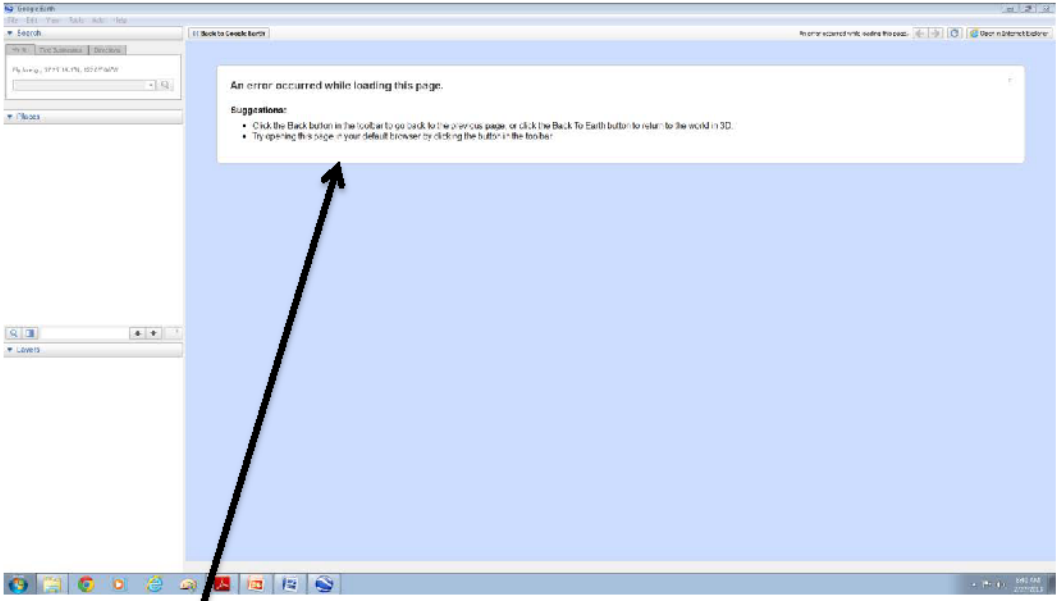






9. Help Users Recognize, Diagnose, and Recover from Errors

If a user enters an address that is not in the maps database, Google maps will suggest alternatives to help the user figure out the correct address they are looking for.

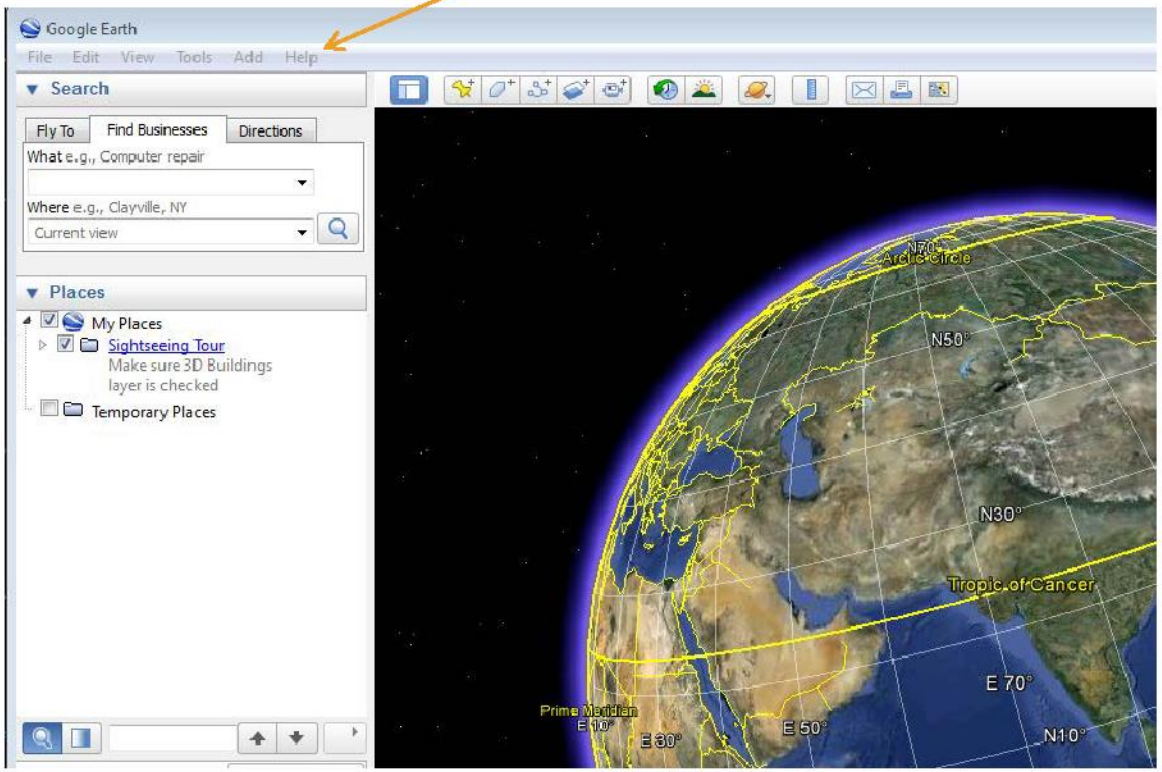


The software is user friendly.
It does not penalise the user for not knowing how to use the software.

Even if an error happens, the software recovers quickly without the user having to do much more than going back to a previous screen by pressing " Back" button.

10. Help and Documentation

In the Documentation under ‘ Help’ it is easy to use and figure out where to find the information you're looking for.



Conclusion

The Heuristic evaluation is compiled into a consolidated report by including results of other evaluators. Intensity of the problem may also be indicated in terms of severity. High severity means it is a HCI problem. Medium means the problem needs attention as it is partially resolved. Low means improvement can still be done to the existing state.

Heuristics	Evaluator 1		Evaluator 2		Evaluator 3		Evaluator 4	
1. Visibility of System Status	**System status if the Network connection is lost is absent**	Severity : Medium						
2.Match between System and Real World	**Good. No intervention required**	NA						
3. User Control and Freedom	**For a novice user disappearing zoom slider bar can be confusing!**	Severity : Low						
4. Consistency & Standards	**Good. No intervention required**							
5. **5. Error Prevention**	**Good. No intervention required**							
6. Recognition Rather than Recall	**Navigating by panning and zooming often leads to being lost . Direction of movement is required**	Severity : HIGH						

Topic for Presentaion-2

- Choose any interface of a device or a website and conduct an audit to identify where Nielsen's ten heuristics have been
  - Adhered to
  - Not adhered to
  - Suggest correction if any?